

ANIA

Assessment Report

**Annex 1 - Analytical Foundations and
Methodological Reasoning**

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BEYOND HORIZONS

- Space Exploration
- New Frontiers
- Knowledge for All

EMPOWERING PEOPLE

- Better Education
- Better Healthcare
- Equal Opportunities
- Stronger Together

SMART & SUSTAINABLE LIVING

- Clean Energy
- Smart Mobility
- Healthy Planet
- Connected Communities

COLLABORATION AT WORK

- Human + AI
- Safer Workplaces
- New Skills
- Shared Success

SMART MANUFACTURING HUMAN + AI COLLABORATION

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Introduction

This Annex documents the analytical and methodological foundations on which the ANIA Assessment Report rests:

- Part A defends the indicator framework that grounds the comparative claims in §4 of the main body. The four-category decomposition (Jobs, Democracy, Social cohesion, Ethics and Human Development), the 6 criteria governing pillar admission, the 20 pillars themselves with their rationale and 3 levels scoring criteria, and the 23 countries dataset from which the eight comparators discussed in §4 were drawn.
- Part B presents the two strategic-synthesis instruments, SWOT and PESTLE, that bridge §4 and the targeted recommendations in §5.
- Part C describes the ANIA Application, the interactive companion application that exposes the dataset directly to readers and examiners through a map-and-chat interface.

Part A. Methodology

A.1 The concept of social cohesion adopted

Social cohesion is operationalised in the report as the *quality of horizontal and vertical relations* between citizens and between citizens and the institutions that govern them, following Schirch (2023). The horizontal dimension covers trust, cooperation, and mutual recognition across groups within the citizenry, lines along which AI deployment in public services, hiring, credit, and welfare can either ease or sharpen division. The vertical dimension covers the relationship between citizens and the institutions that act on them, the locus at which AI in public administration is most consequential because it changes the texture of the citizen-state encounter without necessarily changing the law that governs it.

The choice of this relational definition over a stock-of-social-capital framing (associated with Putnam) is deliberate. A relational definition is operationally tractable for assessing a national AI strategy. Every named initiative in such a strategy can be examined for whether it expands or contracts citizen agency, intergroup trust, or institutional legitimacy. A stock-of-capital framing asks a question (does this strategy raise or lower aggregate social capital?) that no national AI agenda is designed to answer and that no available measurement instrument can resolve at the level of a published strategy text. The relational definition produces tractable comparative claims at the level of named initiatives; the stock framing would have produced unresolvable disputes about national aggregates.

The report also draws on Therborn's (2009) *distantiation* mechanism: front-runners pulling ahead of the rest without explicit barriers being erected, the gap widening through differential access to capability instead of through any single discriminatory act. *Distantiation* names the cohesion harm that an AI strategy can do quietly, by accelerating capability for one cohort while leaving another behind, even when no provision in the strategy explicitly excludes anyone. The lens matters because it forces the assessment to look beyond what an AI agenda includes, to the asymmetries its successful execution would generate.

Those asymmetries decide whether the agenda raises the floor or merely raises the ceiling, and that decision is invisible to any framework that scores only the presence of provisions.

A.2 Pillar-selection methodology

Within each of the four course-given categories, the report decomposes the category into five sub-indicator pillars. A candidate pillar was admitted into the framework only if it satisfied all six of the following criteria.

The first criterion is **observability in the strategy text**. A pillar must be codeable directly from the published national AI strategy or plan, without inferring unstated intent. The criterion bars any pillar whose score would depend on tacit interpretation of what a strategy "really means" beyond what its text says. It accepts that a strategy may have

unwritten ambitions and that implementation may diverge from the document; what it refuses is to score the unwritten or the unimplemented.

The second criterion is **operational-gradient discrimination**. A pillar must support the three-level rubric, described in A.4, that separates a stated principle from a named commitment from an operational or quantified instrument. Pillars on which every agenda in the corpus would score identically were rejected as non-discriminating, because they would carry no comparative signal and would add only nominal coverage.

The third criterion is **cross-national comparability**. A pillar must be meaningful across agendas that differ widely in regime, size, language, and maturity. Indicators specific to one regulatory tradition (for example, scoring strict EU-AI-Act conformity) would systematically advantage one cluster of countries and penalise another for reasons unrelated to the pillar's substantive content, and were excluded on that basis. Where a pillar nevertheless touches the EU AI Act (the Ethics-and-Human-Development pillar on AI Act or equivalent), the rubric is written to credit any dedicated AI statute, not only the EU's, so the comparability criterion holds.

The fourth criterion is **non-redundancy**. A pillar must isolate a mechanism that the other pillars in its category do not already capture. Double-counting would inflate the composite scores of agendas that happened to formulate a single commitment in language triggering multiple pillars, and would deflate composites for agendas that bundled the same substance into fewer textual moves. The criterion is also why the framework documents specific cross-category overlap cases (notably the *sovereign LLM* pillar in Ethics-and-Human-Development and the *linguistic substrate* pillar in Social cohesion) instead of pretending the overlap does not exist.

The fifth criterion is **cohesion-materiality**. A pillar must be a concrete lever through which the category bears on social cohesion, the project's focal lens. Indicators that captured generic AI-strategy quality but had no plausible cohesion mechanism (raw research-publication volume being the obvious case) were excluded, because they would have widened the framework's scope away from the assessment's stated purpose without yielding distributional information.

The sixth criterion is **parsimonious coverage**. The set is held to five pillars per category. Five is enough to span the principal cohesion-relevant dimensions of each category without fragmenting into noise; more would have generated low-information sub-pillars that diluted the signal, fewer would have left observable cohesion mechanisms uncoded. The number is a deliberate methodological judgement, defended on its own terms, not a constraint imported from elsewhere.

The set of five pillars per category was constructed to be MECE-leaning: mutually exclusive (each pillar codes a different mechanism) and collectively exhaustive (the five together aim to cover the category's principal cohesion-relevant surface). Strict MECE is not claimed. Some interaction across pillars within a category is unavoidable, particularly where a single national instrument speaks to more than one mechanism at once. The framework therefore documents the principal overlap cases transparently, both within and across categories, so that readers can apply their own correction if they judge a

country's composite score to have been pushed up or down by a textual artefact of how its strategy is written.

A.3 The scoring rubric

The sub-indicator rubric follows the same three-level progression across nearly every pillar in the framework. A score of **0** records the absence of the pillar's substance, or its presence at principle level only. A score of **1** records a named or announced commitment, present rhetorically in the strategy text but not yet operationalised, quantified, or institutionalised. A score of **2** records an operational, quantified, institutionalised, or mandated instrument: a register that exists, a target with a number attached, a body with seats assigned, or a statute with binding force.

The rubric encodes a single substantive criterion: *how far a commitment has moved beyond words*. It does not measure the moral attractiveness of a commitment, nor its scale, nor its alignment with any regulatory tradition.

A strategy that announces a fairness toolkit scores 1 on the corresponding pillar; a strategy that operationalises an open-source fairness toolkit scores 2, irrespective of which toolkit it is and irrespective of whether the assessor considers the toolkit politically attractive. A strategy that allocates seats on an ethics body to civil society scores 2 on the corresponding pillar; a strategy that mentions civil-society participation without seats scores 1.

Score	Meaning
0	Absent, or principle-level only
1	Named or announced (rhetoric, not yet operationalised)
2	Operational, quantified, institutionalised, or mandated

The category-level composite score is bounded at **0-3** instead of at 0-10 (the simple sum of five 0-2 pillars). The 0-3 banding is interpretive: it reflects the assessor's reading of how operationally serious the category is in the agenda as a whole, anchored to the underlying five pillars but not mechanically derived from them. A 0 composite indicates a category effectively absent from the agenda's operational substance; a 3 indicates a category that the agenda treats as central, with multiple operational pillars supporting it.

The composite score is therefore the load-bearing comparative claim, and the five underlying pillars are the auditable breakdown that allows a reader to inspect how the composite was reached. Where pillar scores and composite would suggest different readings, the rubric in A.5 lets a reader re-derive a different composite if they disagree with the assessor's interpretive weighting.

A.4 The twenty pillars

The four category blocks below set out the five pillars admitted into each category, with a one-line rationale and the three-level criteria. The wording of the criteria is the operational form of the rubric in A.4 specialised to each pillar's substantive content.

A.4.1 Jobs

The Jobs category asks how an agenda translates AI-driven labour-market disruption into concrete workforce, representation, and productivity commitments: who is trained, who has a voice in deciding, who is protected, and what is promised.

Workforce training cohorts. Reskilling is the most common labour pledge in national AI agendas; the test is whether it is sized. The pillar distinguishes a vague training promise from a measurable commitment to a defined cohort. 0: no workforce training target. 1: cohort named without numbers. 2: numeric cohort target.

Specialist pipeline. Captures whether the agenda builds high-end AI talent (researchers, engineers, applied scientists) and whether that ambition is quantified instead of merely asserted. 0: no specialist track. 1: specialist track named. 2: quantified specialist target.

Union or social dialogue. Indicates whether workers have structural voice in AI governance, which is a cohesion signal, instead of being consulted symbolically. 0: not addressed. 1: consultative reference. 2: institutional representation in governance.

Distributional monitoring. Tests whether the agenda watches who wins and who loses from AI-driven productivity, with operational monitoring instead of aspirational language. 0: not addressed. 1: distributional language without monitoring. 2: operational monitoring or binding target.

Productivity target (quantified). Reveals whether economic gains from AI are stated as an auditable headline figure or remain a generic ambition. 0: no headline productivity figure. 1: qualitative productivity ambition. 2: quantified productivity or GDP-add figure.

A.4.2 Democracy

The Democracy category asks whether the agenda embeds democratic safeguards (participation, oversight, transparency, information-ecosystem defence) into how AI is governed and deployed by the state.

Civic participation. Measures whether citizens shape AI policy by design, not only after the fact: moving from announced consultation to institutionalised participation. 0: no civic participation provision. 1: consultation announced. 2: institutionalised participation in design.

Multi-watchdog oversight. Plural, independent oversight guards against capture; the test is whether a watchdog mechanism actually operates on a regular cadence instead of being merely announced. 0: no multi-watchdog mechanism. 1: watchdog body announced. 2: operational with cadence.

Civil society in ethics body. Checks whether civil society holds mandated seats in the operational ethics body, instead of being referenced without real representation. 0: civil society not represented. 1: mention without seats. 2: mandated representation in operational body.

Transparency register. A public register of state AI use is the operational backbone of accountability; the pillar distinguishes a stated transparency principle from a working register. *0*: no transparency commitment for public AI. *1*: transparency principle stated. *2*: operational register of public AI use.

Election or disinformation instruments. Tests whether the agenda confronts AI threats to the information ecosystem (elections, disinformation) with concrete instruments instead of principle-level mentions. *0*: not addressed. *1*: mentioned at principle level. *2*: operational instrument. The top of the 0-2 scale is not reached by any country in the dataset, an empirical finding that the rubric captures rather than hides.

A.4.3 Social cohesion

The Social cohesion category is the report's focal lens: it asks whether AI policy actively works to hold society together (across groups, languages, and territory) instead of treating cohesion as a derivative effect.

Structural pillar. Captures how central cohesion is to the agenda's architecture: a guiding principle only, a sub-pillar inside an Area, or a central or near-central operational concern. *0*: cohesion in guiding principles only. *1*: sub-pillar in some area. *2*: central or near-central operational concern.

Algorithmic fairness toolkit. Fairness needs tooling, not only values; the pillar tests whether a concrete (ideally open-source) fairness instrument is operational instead of stated as principle. *0*: principles only. *1*: instrument announced. *2*: operational open-source toolkit.

Linguistic substrate. Sovereign-language models prevent linguistic exclusion; the strongest form pairs a sovereign model with sectoral data instead of merely naming the model. *0*: no sovereign-language model. *1*: sovereign model named only. *2*: sovereign model paired with sectoral data.

Territorial distribution. Guards against capital-centric AI; the pillar measures whether capacity is distributed across regions through an operational hub network, instead of being nationally anchored at a single centre. *0*: nationally anchored. *1*: regional distribution announced. *2*: distributed hub network operational.

Digital inclusion cohorts. Tests whether inclusion of marginalised populations is backed by numeric coverage targets instead of qualitative goodwill. *0*: no cohort coverage target. *1*: qualitative coverage commitment. *2*: numeric cohort targets including marginal population.

A.4.4 Ethics and Human Development

The Ethics-and-Human-Development category asks whether the agenda anchors AI in sovereign capability, openness, enforceable rules, working ethics institutions, and an explicit human-development orientation.

Sovereign LLM. Sovereign model capability underpins autonomy over AI; the pillar distinguishes an announced model from one actually operational. 0: no sovereign LLM. 1: sovereign model announced. 2: sovereign model operational.

Open-source commitment. Openness affects scrutiny and equitable access; the strongest form mandates open-source or open-weight distribution instead of mentioning it in principle. 0: no open-source commitment. 1: open-source mentioned in principle. 2: mandated open-source or open-weight.

AI Act or equivalent. Enforceable rules matter more than guidance; the pillar tests for a dedicated AI statute (or EU AI Act anchoring) versus sector-based or absent regulation. 0: no AI-specific statute or regulation. 1: sector-based without dedicated statute. 2: EU AI Act anchoring or dedicated AI statute.

Ethics body operational. An ethics body is meaningful only if it works and includes outside voices; the pillar distinguishes an announced body from one operational with civil-society participation. 0: no ethics body. 1: ethics body announced. 2: operational with civil-society participation.

Human-development alignment. Tests whether human development (the Human Development Index framing, broadly) is a central metric of the agenda instead of being unmentioned or merely referenced. 0: HDI or human-development framework not named. 1: referenced. 2: central metric or framing.

A.5 The 23 countries corpus (and the 8 chosen derivation)

The corpus comprises twenty-three national AI agendas, chosen across all inhabited regions of the world to span the design space within which any individual country, including Portugal, can be read. The selection rule for the corpus itself is the inverse of the selection rule applied inside §4 of the main body: the corpus aims for *breadth* of regime, size, language, and AI-strategy maturity, so that the indicator distribution against which Portugal is benchmarked is heterogeneous instead of European-and-OECD-bound.

The full list, grouped by region, is: Asia-Pacific (Australia, China, India, Japan, New Zealand, Singapore); Europe (Finland, France, Germany, Italy, Portugal, Spain, Sweden, United Kingdom); Americas (Argentina, Brazil, United States); Africa (Egypt, Kenya, Nigeria, South Africa); Middle East (Qatar, UAE).

The narrowing from twenty-three to eight, defended at length in §4.1 of the main body, applied a different rule: deliberate paired-contrast within each of four world regions, two comparators per pair, staged so that each pair answers a different version of the same question (how does the strategy organise itself when social cohesion is the lens?).

The corpus is therefore the *quantitative* backbone of the report (it sustains the rank positions, the cross-national distributional claims, and the SWOT and PESTLE positioning in Part B), while the chosen eight are the *qualitative* selection inside which the report develops its substantive argument. The two layers are complementary: the corpus says where Portugal sits among twenty-two peers; the chosen eight say what Portugal could learn, organisationally, from a curated set of contrasting strategies.

A **scope-exception protocol** governs the few cases where the report draws on a country outside the chosen eight. The principal case is Singapore's *AI Verify*, cited in §5.2 of the main body as the open-source algorithmic-fairness toolkit available to the Centre for Responsible AI. The exception is principled, not ad hoc: no country in the chosen eight supplies an equivalent open-source instrument, and a recommendation requires the reader to know what such an instrument looks like in practice.

The rule for any future scope-exception is the same: identify the instrument needed, confirm that no chosen comparator supplies it, and surface the exception transparently in-text instead of in a footnote. The corpus is what makes the protocol auditable, because it documents which alternatives were considered and rejected.

A.6 Limits and reflexivity

The framework's limits should be named explicitly so that the reader can locate the assessment's reliability boundary.

Single-coder rating. The scoring across twenty-three countries and twenty-four indicators was carried out by a single assessor. No inter-rater agreement check was performed. The rubric in A.4 and the per-pillar criteria in A.5 are designed to make the scores reproducible in principle, but the absence of an actual second coding is a real limit. The detailed three-level criteria, combined with the source-document traceability of every score (the underlying per-country research summaries that grounded each cell), are what the report offers in place of an inter-rater statistic.

Snapshot in time. The matrix is dated June 2026 and captures each agenda at the moment of its most recent publication. ANIA is read at Resolution n.º 2/2026; the other comparators are read at their publication dates as listed in the References. National AI agendas are revised on cycles of two to five years, so the matrix will need refresh as a matter of course. The 2027 mid-cycle review proposed in §5.3 of the main body would be the natural Portuguese trigger for a re-scoring.

Strategy-text-only evidence base. The framework scores what the strategy text commits to, not what the strategy has yet implemented in practice. A 2 on a pillar therefore means that the strategy text records an operational, quantified, institutionalised, or mandated instrument; it does not certify that the instrument is functioning as written. The framing is appropriate to the comparative question the report asks (how do national AI agendas position themselves on social cohesion at the moment of writing?), but it is not appropriate to an implementation audit, which would require fieldwork, interviews, and access to internal-government data the assessment did not pursue.

Two-decomposition issue. The main body's §2 uses a five-sub-dimension lens for the deep ANIA assessment (digital inclusion and AI literacy; algorithmic fairness in public services; labour-market inequality; cultural and linguistic cohesion; regional and territorial cohesion), and the present annex uses a different five-pillar decomposition for the Social cohesion category (structural pillar; algorithmic fairness toolkit; linguistic substrate; territorial distribution; digital inclusion cohorts). Four of the five §2 sub-dimensions map cleanly onto §A.5 cohesion pillars (inclusion to inclusion cohorts; fairness to fairness

toolkit; linguistic to linguistic substrate; regional to territorial distribution). The fifth §2 sub-dimension, labour-market inequality, sits in the Jobs category in the present annex, not in Social cohesion. The split is principled (labour-market inequality is the cohesion mechanism that the Jobs category indicators are designed to capture, particularly the union-or-social-dialogue and distributional-monitoring pillars), but it is also a place where a reader can reasonably want the framework to be more consolidated. The report's choice was to keep the §2 lens optimised for narrative depth and the present annex's lens optimised for cross-national comparability; the cost of the choice is documented here so the reader can re-weight either lens as preferred.

Interpretive composite. The 0-3 composite is interpretive, not mechanically derived from the five 0-2 pillars. A reader who disagrees with the assessor's banding can re-derive a composite from the pillar scores using a different rule (sum, mean, threshold count) without altering the underlying data. The pillars are the auditable layer; the composite is the comparative claim.

Part B. Strategic synthesis tools

Two synthesis instruments translate the indicator-level evidence in Part A and the country narrative in §4 of the main body into a form usable by the recommendations in §5:

- **SWOT** organises the assessment around Portugal: strengths and weaknesses internal to ANIA and the Portuguese policy stack (§2), opportunities and threats external (§4 and the wider EU framing).
- **PESTLE** organizes the assessment around the comparator set: it positions Portugal alongside the eight chosen comparators across six dimensions (political, economic, social, technological, legal, environmental), showing for each cell where Portugal sits and what the contrast reveals. Read together they bridge the inward-facing diagnostic and the outward-facing benchmark.

B.1 SWOT analysis of Portugal's social-cohesion AI position

The SWOT below synthesises the report's findings into the four canonical quadrants. Strengths and Weaknesses are internal to ANIA and to the Portuguese policy stack and draw on §2 of the main body. Opportunities and Threats are external and draw on the eight-country comparator analysis in §4 and on the wider EU framing in §3. The cells are stated tersely; the rationale for each lives in the corresponding section of the main body.

Table B.1 - SWOT analysis of Portugal's social-cohesion AI position.

Internal (ANIA and the Portuguese policy stack)		External (EU framing and comparator strategies)	
Helpful	Strengths S1. Coherent EU AI Act regulatory perimeter anchored in Area IV (IV.3, IV.4, IV.6). S2. Named sovereign Portuguese-language model (AMALIA, II.7). S3. Explicit citizen-AI-literacy initiative (III.8) with cross-ministerial leadership (ANACOM, CNCS, IPQ, ARTE). S4. Embedded in the broader EDN, providing a territorial-cohesion backbone (Action 11.4 Mobile Citizen Service Points; Action 13.4 CAM/Azores submarine cables). S5. Centre for Responsible AI (IV.3) as institutional locus for ethical, social, and economic impact assessment.		Opportunities O1. French Humanism pillar and Finnish strategic-level roundtable as transferable templates for a Portuguese AI Observatory with tripartite social-dialogue representation. O2. Australian multi-watchdog AI Review Committee as a low-cost mechanism for closing the algorithmic-fairness operational gap. O3. Singaporean <i>AI Verify</i> open-source toolkit, adoptable by the Centre for Responsible AI at near-zero direct cost (scope-exception reference). O4. Bilateral AMALIA coordination with Brazilian PT-BR alongside the Egyptian paired model-and-substrate approach, addressing the linguistic-cohesion substrate gap. O5. Brazilian <i>AI for the Good of All</i> and Egyptian <i>Inclusive AI</i> framings as templates for a Society and Inclusion cross-cutting sub-pillar within Area IV. O6. South African AIISA distributed-hub model (eleven hubs by 2025) as a template for a National AI Hub Network with autonomous-region coverage. O7. Brazilian 50% women-researcher target and India's Jan Dhan gender mainstreaming (490 million informal workers as named target) as operational templates for cohort quotas and inclusive scope. O8. Apply AI Strategy §3.2 commitment to “address structural inequalities”, a direct EU alignment opportunity.
Harmful	Weaknesses W1. No structural Society pillar: cohesion is folded into guiding principles, under-operationalised relative to seven of the eight chosen comparators (only the United States similarly treats cohesion as a derivative effect). W2. No quantified per-cohort annual education targets (contrast Brazil, India, South Africa, Egypt). W3. No central register of AI impact assessments, no multi-watchdog committee, no AI Ethics Expert Group, no algorithmic-transparency standard. W4. Doutor AP (III.1) addresses an advanced cohort instead of the universal civil-servant base (contrast Australia). W5. No commitment to monitoring AI's distributional effects across gender, age, disability, or migrant origin. W6. AMALIA is not paired with a sectoral Portuguese data-sets initiative (contrast Egypt) and is not committed to open-source or open-weight distribution. W7. No ANIA-internal regional-cohesion provision and no distributed AI-hub network (contrast South Africa AIISA). W8. No measurement architecture analogous to Egypt's 89-indicator Capacity Assessment Framework; no public AI Cohesion Monitor.		Threats T1. Without distributional monitoring, the <i>distantiation</i> mechanism (Therborn, 2009), where front-runners pull ahead while others fall behind without explicit barriers, remains invisible to ANIA. T2. Anglophone foundation models dominate the global market and embed Anglophone cultural categories into every downstream application (Castells, 2024); AMALIA cannot counter this alone without bilateral coordination with Brazilian PT-BR and a paired Portuguese data substrate. T3. Algorithmic management spreads through Portuguese workplaces without union safeguards comparable to French social dialogue or Finnish trade-union representation. T4. EU peer Member States already operationalize monitoring, transparency, and inclusion commitments, with risk that Portugal falls behind in cohesion-relevant EU indicators. T5. Without explicit territorial provisions inside ANIA, the centre-periphery digital divide accelerates as AI deployment concentrates in Lisbon and Porto; the grey divide (40% of 65 to 74 not using Internet) is the baseline at risk. T6. The Dutch childcare benefits scandal, in which 26,000 mostly immigrant-background families were wrongly accused of fraud by a biased algorithm (FRA, 2022), is the cautionary case for public-administration AI without operational safeguards.

Interpretation. The strengths concentrate on regulatory anchoring and a small set of named initiatives (AMALIA, the Centre for Responsible AI, the citizen-literacy action), which are the assets ANIA already owns and which the recommendations in §5 are designed to build instead of replace.

The weaknesses concentrate on the absence of named operational instruments (registers, committees, quotas, hub networks, measurement frameworks) that seven of the eight chosen comparators articulate explicitly. The contrast between the two internal columns is the report's central diagnostic: ANIA's regulatory and infrastructural foundations are solid, and its operational vocabulary of social-cohesion instruments is thin.

The opportunities cluster around eight directly transferable instruments anchored to existing ANIA initiatives or EDN Actions, drawn from the chosen comparators with one scope-exception (Singapore's *AI Verify*) where no chosen country supplies an equivalent open-source instrument.

The threats are temporal: they grow if cohesion-relevant monitoring and operational instruments are not put in place within the 2026 to 2030 horizon. The nine targeted recommendations in §5 convert the opportunities into operational commitments and neutralise the principal threats, positioning Portugal at the intersection of the European Union, the Lusophone world, and the African continent.

B.2 PESTLE positioning of Portugal and the 8 chosen comparators

The PESTLE below positions Portugal alongside the eight chosen comparators across the six PESTLE dimensions. Each cell is anchored to the strategy document or programme identified in §4 of the main body and in the per-country research summaries. The table is written for at-a-glance comparison; the cross-cutting interpretation that follows draws out the patterns that matter for ANIA's positioning.

Table B.2 - PESTLE positioning of Portugal and the eight chosen comparators.

Country	Political (P)	Economic (E)	Social (S)	Technological (T)	Legal (L)	Environmental (E)
Portugal (<i>ANIA 2026 and EDN PA 2026-27</i>)	Centre-right PSD/CDS; ANIA = Action 20 of EDN; ARTE (Decree-Law 96/2025) operational lead; EU and CPLP diplomatic priorities.	Headline claim of €18-22 B/yr GDP add and 2.7 pp productivity uplift; 3% national R&D target; no per-initiative budget itemized.	Six guiding principles, with cohesion folded into principles instead of being operationalized; weak worker-voice mechanism.	AMALIA sovereign Portuguese-language model (II.7); Deucalion HPC (~5% PT quota); Centre for Responsible AI as advisory locus.	EU AI Act perimeter alongside GDPR and Lei 27/2021; sectoral oversight via CNPD and ANACOM; no dedicated national AI Act.	~60% renewable electricity; submarine-cable hub; sustainable-AI commitments implicit.
Australia and New Zealand (<i>National AI Plan Dec 2025; AI Strategy Jul 2025</i>)	Australia: Labor (Albanese), “shared prosperity”; cross-ministerial NAP and APS plan. New Zealand: National-led, MBIE-driven, with a separate public-sector track under the Minister for Digitizing Government. Both common-law and indigenous rights anchored.	Australia: A\$116 B GDP add and 4.3 pp productivity by 2030; A\$460 M existing, A\$1 B NRF, and A\$950 M RDTI. New Zealand: projected NZ\$76 B GDP add by 2038 (15% of GDP); Budget 2025 NZ\$388 M total tertiary AI funding.	Australia: First Nations Data Sovereignty named; union-consultation mandated; APS foundational training universal within 12 months. New Zealand: Treaty of Waitangi embedded; <i>mātauranga Māori</i> treated as <i>taonga</i> ; 68% of SMEs have no plans to adopt AI.	Australia: GovAI vendor-agnostic platform; National AI Sandbox; AISI under NAP Action 7. New Zealand: explicit adoption focus (no foundational LLM); Catalyst Fund AI priority; WEKA (1996) open-source heritage.	Australia: sector-based and tech-neutral, <i>not</i> the EU AI Act path; AI Review Committee multi-watchdog. New Zealand: light-touch and principles-based; OECD AI Principles adopted in explicit rejection of an AI-Act equivalent; Algorithm Charter for Aotearoa.	Australia: 100% net-zero-electricity DC principles. New Zealand: ~80% renewable electricity; rural and agricultural sustainability via sectoral AI deployment.
India (<i>IndiaAI Mission, Cabinet 7 Mar 2024</i>)	BJP-led (Modi); MeitY and NITI Aayog implementation; #AIForAll vision; co-chair of AI Action Summit Paris 2025; GPAI founding member; India AI Impact Summit 2026 host.	~US\$1.25 B (₹10,371 crore) over 5 years; ₹2,000 crore Union Budget 2025-26 allocation; aligned with Viksit Bharat@2047; demographic dividend until 2055-56.	Largest cohesion ambition in the corpus: village-assembly AI (SabhaSaar in 14 languages); Jan Dhan gender mainstreaming (530 M accounts, 55% women); 490 M informal workers as named policy target.	BHASHINI (36+ languages, 350+ models, 23+ government services); BharatGen sovereign LLM (22 languages); 18,000+ GPUs subsidized at ~US\$0.78/hour; AIRAWAT supercomputing.	DPDP Act 2023; India AI Governance Guidelines (MeitY 2025) with seven <i>Sutras</i> across six pillars; Cabinet-approved mission with no parliamentary statute (structural fragility).	Mixed coal and expanding renewables; rural climate-resilience AI deployment; no aggregate environmental-AI framing at strategy level.

Country	Political (P)	Economic (E)	Social (S)	Technological (T)	Legal (L)	Environmental (E)
France (<i>Stratégie nationale 2024</i>)	Macron / Renaissance; per-recommendation ministry assignment; Germany priority and Africa bilateral focus; AI Action Summit host (Feb 2025); EU and Francophonie diplomatic anchoring.	€27 B/5 yr value-creation goal through 2029; <i>Conseil d'analyse économique</i> backing; top-tier EU publications.	Three-pillar architecture (Humanism, Sovereignty, Responsibility) with Humanism first; <i>AI Cafés</i> as citizen-deliberation spaces; explicit social-dialogue mandate; lifelong learning across pillars.	Jean Zay and Adastra HPC; Mistral and Bull/Atos sovereign-compute stack; INRIA and ANSSI national institutes.	EU AI Act, GDPR, and <i>Loi Informatique et Libertés</i> ; EU CDSM; CNIL oversight; no national AI Act.	Nuclear-dominant low-carbon electricity (~70%); EU Green Deal alignment; sectoral DC sustainability.
Finland (<i>AI 4.0 Programme 2022</i>)	Cross-coalition continuity (Rinne, Marin, Orpo era); TEM (Ministry of Economic Affairs and Employment) Steering Group; EU and Nordic priority; “Twin transition and carbon handprint” positioning.	TEM-funded, not aggregated; 10-year strategy with 2027 milestones; aligned with EU 3% R&D target.	Strong civic dialogue; <i>Elements of AI</i> (originated in Finland, global influence); twin-transition and youth focus; Nordic worker-voice in roundtable governance.	LUMI HPC (EuroHPC); no sovereign LLM; EU Data Spaces; EU-aligned standards.	EU AI Act, GDPR, EU CDSM; Data Protection Ombudsman oversight; quantitative international-ranking targets as performance discipline.	Strong renewables (75%+); cold-climate cooling advantage; <i>Twin transition and carbon handprint</i> framing as a distinctive metric; energy-efficient AI focus.
United States (<i>America’s AI Action Plan, July 2025</i>)	Trump second-term Republican; White House OSTP; partisan single-administration framing; Indo-Pacific and UK bilateral; “Frontier dominance” positioning.	No headline dollar target; through 2030 horizon; 1 globally in publications and patents; frontier-leadership goal.	Worker-first reskilling agenda; §132 IRC tax-exempt AI-training incentive; Labor Market Analysis Hub. DEI references actively <i>removed</i> from NIST AI RMF and from federal procurement guidance.	Frontier private sector (OpenAI, Anthropic); NIST AISI; FedRAMP federal cloud; OMB procurement guidance for federal LLM use.	Sector-based (explicit anti-EU AI Act stance); CCPA, HIPAA; executive orders instead of statute; bias references stripped from federal guidance.	Mixed and state-varying; no AI-environmental framing at federal level.

Country	Political (P)	Economic (E)	Social (S)	Technological (T)	Legal (L)	Environmental (E)
Brazil (<i>PBIA 2024-2028, “IA para o Bem de Todos”</i>)	PT social-democratic (Lula); MCTI, CGEE, and CCT; CIT Digital and SGD/MGI; Latin America, Africa, CPLP, and India diplomatic priorities.	R\$23.03 B BRL 2024-28 with 5-axis budget granularity (gold-standard alongside Spain’s <i>Tabla 6</i>); aligned with national S&T strategy.	AI Olympiad; CadÚnico reach to 97 M low-income citizens; cultural, regional, peoples’ diversity, and low-income inclusion explicit; 50% women-researcher target binding.	Top 5 worldwide supercomputer target; PT-BR sovereign LLM (Action 9); National Center for Algorithmic Transparency planned (Action 51); Sovereign Cloud (Action 27).	Bill 2338/23 pending in Congress; LGPD privacy; civil-law liability; ANPD, sectoral, and planned-national-center oversight.	89.2% renewable electricity (2023, highest in corpus); Sustainable AI Pro-Infra (Action 7); N/NE renewable data centers (Action 41); Amazon biome AI deployment.
South Africa (<i>AI Planning, October 2023</i>)	ANC-led coalition government; Department of Communications and Digital Technologies; anchored to National Development Plan 2030; AfCFTA continental data-cooperation orientation.	Investment not itemized at strategy level; 2030 horizon aligned with NDP; BRICS+ alignment.	30% whole-population workforce literacy target; 20,000 AI specialists and 5,000 AI experts; jobs-displacement mitigation explicit; AIISA partnership with universities.	AIISA eleven-hub university network targeted by 2025; deliberately distributed institutional model outside the largest metropolitan center; AfCFTA continental data-space proposal.	Discussion-draft status (Oct 2023), not yet binding; sector-based; POPIA privacy; no dedicated AI legislation.	Mixed coal and expanding renewables; limited environmental-AI framing.
Egypt (<i>National AI Strategy Second Edition 2025-2030</i>)	Authoritative-state context; National Council for AI under MCIT; presidential alignment; African Union and Arab League diplomatic priorities.	Second-edition strategy 2025-2030; not itemised per initiative; explicit composite-score and global-ranking trajectory as performance metric.	26% marginal-population KPI and 36% public AI-access KPI (only quantified vulnerable-population commitment in the corpus); cross-discipline AI literacy; 30,000 AI professionals by 2030.	Arabic-first foundation models (Initiative 3) paired with Sectoral Arabic Data Sets (Initiative 7); ubiquitous 5G and fibre; 89-indicator Capacity Assessment Framework.	Personal Data Protection Law; sector-based; data-sovereignty commitments explicit at strategy level; no AI-specific statute.	Mixed gas and renewables; sustainability-of-DC commitments named under Infrastructure pillar.

Cross-cutting patterns. Several patterns emerge across the chosen eight that matter for Portugal's positioning.

Politically, the comparator set spans coalition stability (Finland, France), populist-frontier framing (United States), social-democratic explicit-inclusion framing (Brazil), and authoritarian developmentalism (Egypt), with Australia and New Zealand sitting between adoption-pragmatism and indigenous-rights anchoring. Portugal occupies a centrist EU-anchored position closest to France and Finland, but without the explicit cohesion-as-purpose framing that distinguishes Brazil and France.

Economically, only Spain (not in the chosen eight) and Brazil match the per-initiative budget granularity that ANIA currently lacks; the Brazilian PBIA's five-axis breakdown is the strongest available template. France's €27 B value-creation goal and Australia's A\$116 B GDP-add headline anchor monetised national-ambition framings of similar weight to Portugal's €18-22 B/year claim. The economic dimension exposes a single concrete weakness in ANIA's structure: a credible headline figure without per-initiative budget lines is difficult to defend in a mid-cycle review.

Socially, the contrast is sharpest between *opportunity-creation* framings (the United States, where cohesion is treated as a derivative effect of workforce reskilling) and *inclusion-as-purpose* framings (Brazil, India, France). India's village-assembly deployment and informal-worker targeting are the deepest social-penetration examples in the set; New Zealand's Treaty of Waitangi embedding and Australia's First Nations Data Sovereignty are the strongest indigenous-rights examples. ANIA's current social column sits closer to the United States than to Brazil, despite Portugal's social-protection tradition: the gap is the W1 weakness in the SWOT and the principal target of the §5 recommendations.

Technologically, the comparators divide between sovereign-LLM strategies (France, Brazil, India, Egypt), pure-adoption strategies (Australia, New Zealand), and hybrid stances (Finland, South Africa). Portugal's AMALIA places it in the sovereign-LLM cluster, but without the substrate-paired data-sets approach that Brazil's Action 9 and Egypt's Initiative 7 demonstrate, which is why AMALIA scores 1 instead of 2 on the linguistic-substrate pillar in §A.7.

Legally, the EU AI Act anchors France, Finland, and Portugal; Brazil's Bill 2338/23 is pending; the United States is explicitly anti-EU-Act; Australia, New Zealand, and South Africa pursue sector-based or principles-based paths without dedicated AI statutes. Portugal's regulatory perimeter is the strongest in the set on this dimension and is the S1 strength on which much of the §5 recommendation set leans for legal cover.

Environmentally, Brazil leads the corpus on renewable share (89.2%), followed by Finland (75%+), New Zealand (~80%), and Portugal (~60%). Finland's *twin transition and carbon handprint* metric is the only explicitly cohesion-relevant environmental framing; Brazil's reservoir-cooling potential and N/NE renewable DCs are the strongest infrastructural examples. Portugal sits mid-cluster and could credibly position as a green data-centre destination, complementing the National DC Plan.

B.3 Reading the two instruments together

SWOT and PESTLE are deliberately asymmetric. SWOT is Portugal-centric and prescriptive: it isolates what ANIA has, what it lacks, what the international environment offers, and what the international environment threatens. PESTLE is comparator-centric and descriptive. It lays out where each chosen comparator sits across six dimensions, with Portugal as the first row, so that the contrasts read structurally instead of as a ranking.

The two instruments cover the same evidence base from different vantage points, and the value of reading them together is the triangulation: a weakness that surfaces in SWOT (for example, W6 on AMALIA not being paired with sectoral data sets) corresponds to a specific PESTLE column (the Technological column for Brazil and Egypt, where the paired approach is documented), and that correspondence is the audit trail that lets the §5 recommendation set be defended as drawn from the evidence instead of asserted on principle.

Where the two instruments disagree (for example, on whether Portugal's regulatory position counts as a strength or as a constraint that limits comparability with non-EU-Act jurisdictions), the disagreement is itself diagnostic and is resolved in §5 by anchoring each recommendation explicitly to an existing ANIA initiative or EDN Action.

Part C. The ANIA Application

C.1 Purpose

The ANIA Application is the interactive companion to this report. It allows exploration of the AI Agendas from 23 countries through a map-and-chat interface. It consists of a Web UI that shows a world choropleth coloured by whichever indicator the reader selects, alongside an AI Assistant’s conversational panel that answers questions in natural language and updates the map in synchrony with each answer.

We have made ANIA Application’s source code open and available in GitHub¹. For added convenience, it is also available online, at <https://logus2k.com/ania>.

C.2 Composite-score visualisation

The first visualisation is a stacked bar chart of the four category composites across all twenty-three countries in the corpus. The x-axis lists the countries in alphabetical order; each country’s bar stacks four segments (Jobs, Democracy, Social cohesion, Ethics and Human Development), each segment sized by the corresponding 0-3 composite score, so that the total bar height for a country can range from 0 to 12.

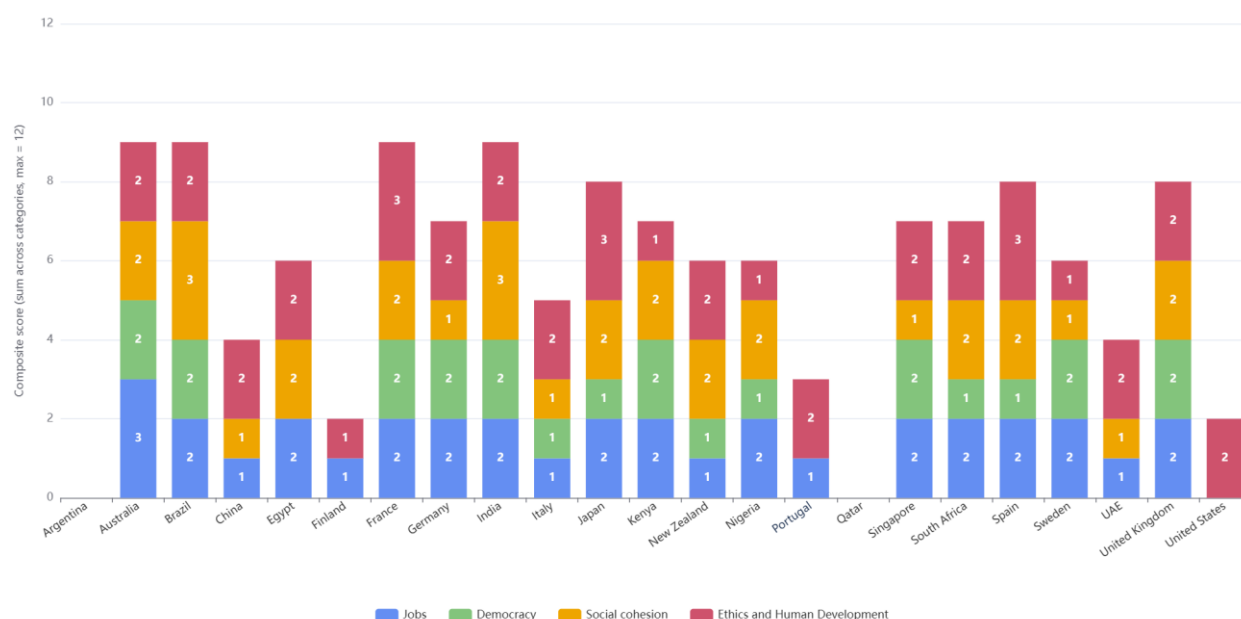


Figure C.2 - ANIA composite scores across the twenty-three-country corpus, by category.

The chart performs two readings at once. The first is the total-height reading: which countries have moved their AI agenda furthest beyond rhetoric across the four categories taken together. France, Spain, Japan, Brazil, India, and the United Kingdom group at the top of the corpus; Argentina, Qatar, and the United States group at the bottom. Portugal sits in the lower-middle band with a total of three, the same total as Finland, Italy, and New Zealand, and below the other Western-European corpus members.

¹ <https://github.com/logus2k/aigeo>

The second reading is the segment-shape reading: where each country's strategic weight is concentrated. Portugal's bar is two-thirds Ethics-and-Human-Development and one-third Jobs, with zero Social cohesion and zero Democracy, a profile shape unlike any other Western-European country in the corpus and one that is the visual confirmation of the central diagnostic in §A.7 and §5.1 of the main body.

C.3 Social-cohesion sub-pillar visualisation

The second visualisation drills one level deeper into the report's focal lens. It is a stacked bar chart of the five social-cohesion sub-pillars (structural pillar, algorithmic fairness toolkit, linguistic substrate, territorial distribution, digital inclusion cohorts) across the same twenty-three countries; each segment is scored 0-2, so the total bar height for a country can range from 0 to 10. The visual style matches the composite chart for consistency; the legend identifies the five pillars in the order in which they are introduced in §A.5.

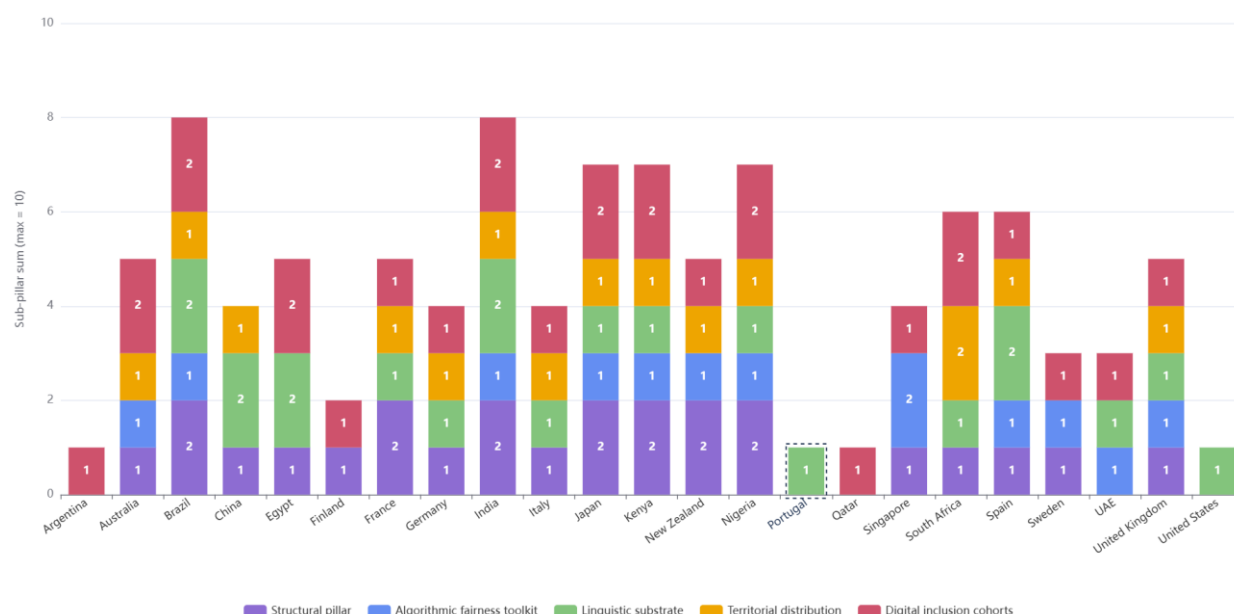


Figure C.3 - Social-cohesion sub-pillar scores across the 23 countries corpus.

The drill-down chart answers a question the composite chart cannot: when a country scores high on the social-cohesion composite, what is doing the work? Brazil and India both sit at composite 3, and the sub-pillar chart shows how: through structural-pillar centrality combined with quantified inclusion cohorts, paired-data linguistic substrate, and operational fairness or territorial distribution mechanisms.

South Africa's social-cohesion composite of 2 is driven by territorial distribution (the only 2 in the column) and digital-inclusion cohorts, with comparatively weaker fairness tooling. Portugal's single point on the entire row, the 1 on linguistic substrate from the named AMALIA model, is visible as the single thin segment in an otherwise empty bar, which is precisely the visual the recommendations in §5.2 are designed to populate over the H2 2026 to H1 2027 windows.

C.4 Jobs, Democracy, Ethics and HD sub-pillar visualization

For completeness, the remaining sub-pillar visualizations were included.

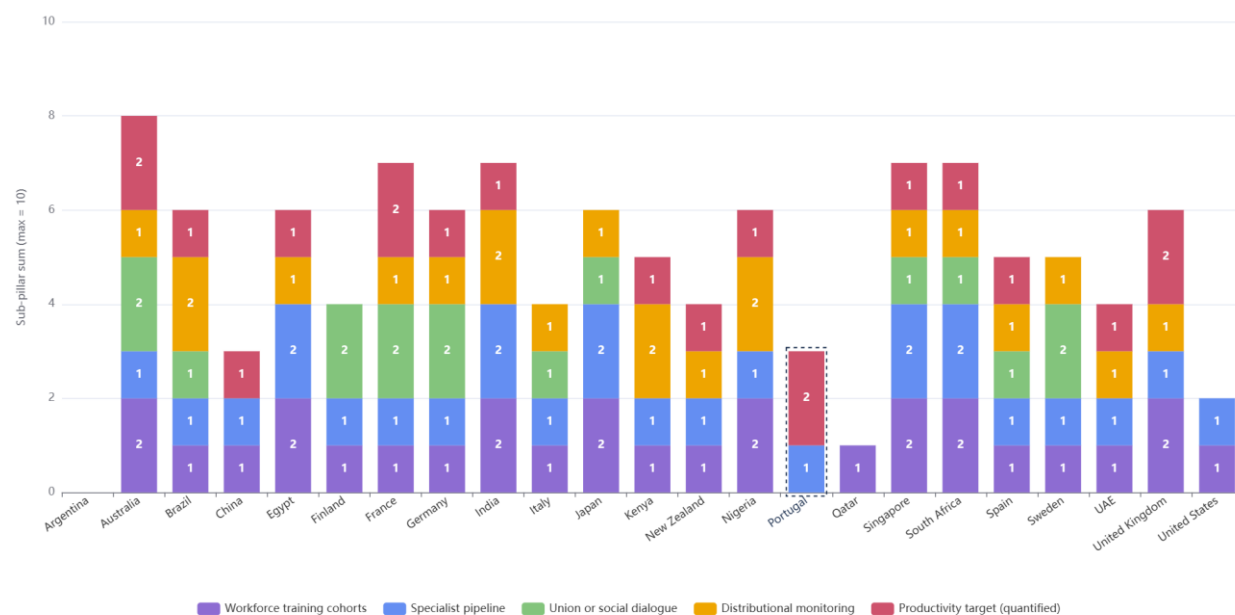


Figure C.4 - Jobs sub-pillar scores across the 23 countries corpus.

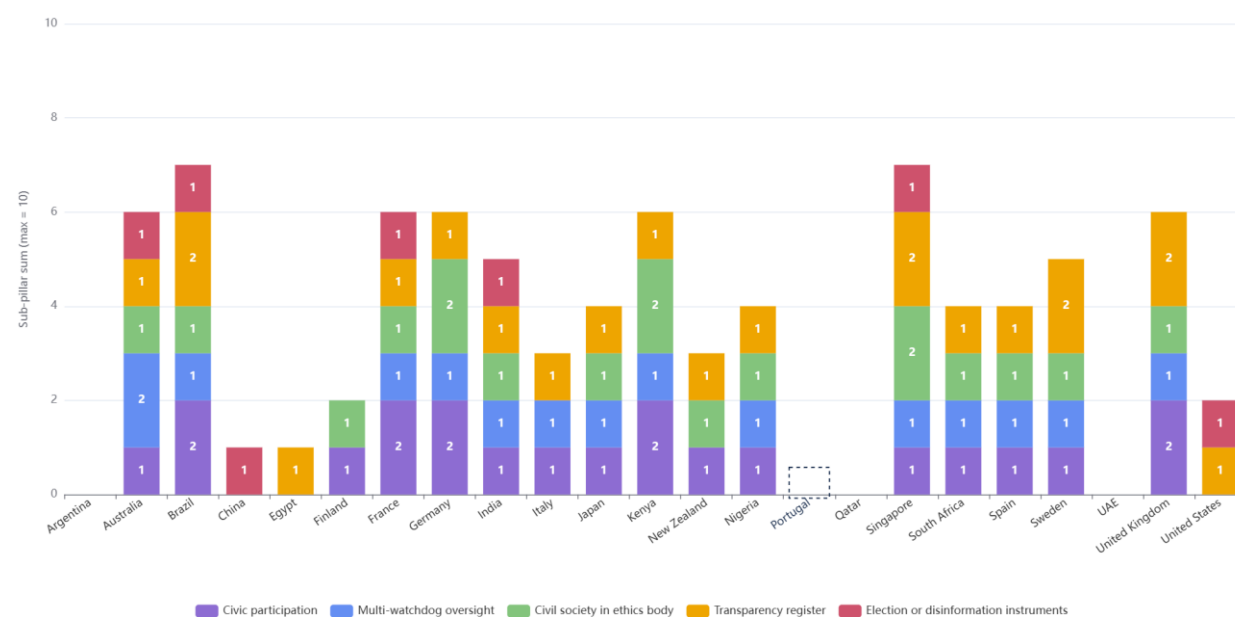


Figure C.5 - Democracy sub-pillar scores across the 23 countries corpus.

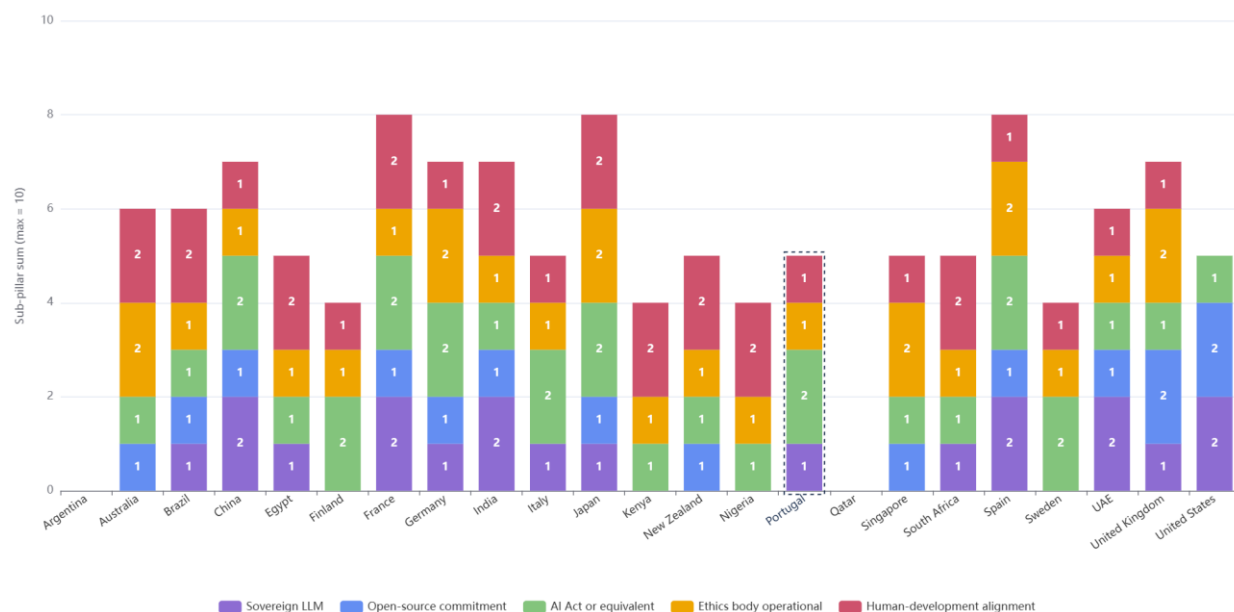


Figure C.6 – Ethics and Human Development sub-pillar scores across the 23 countries corpus.

C.5 Screenshot

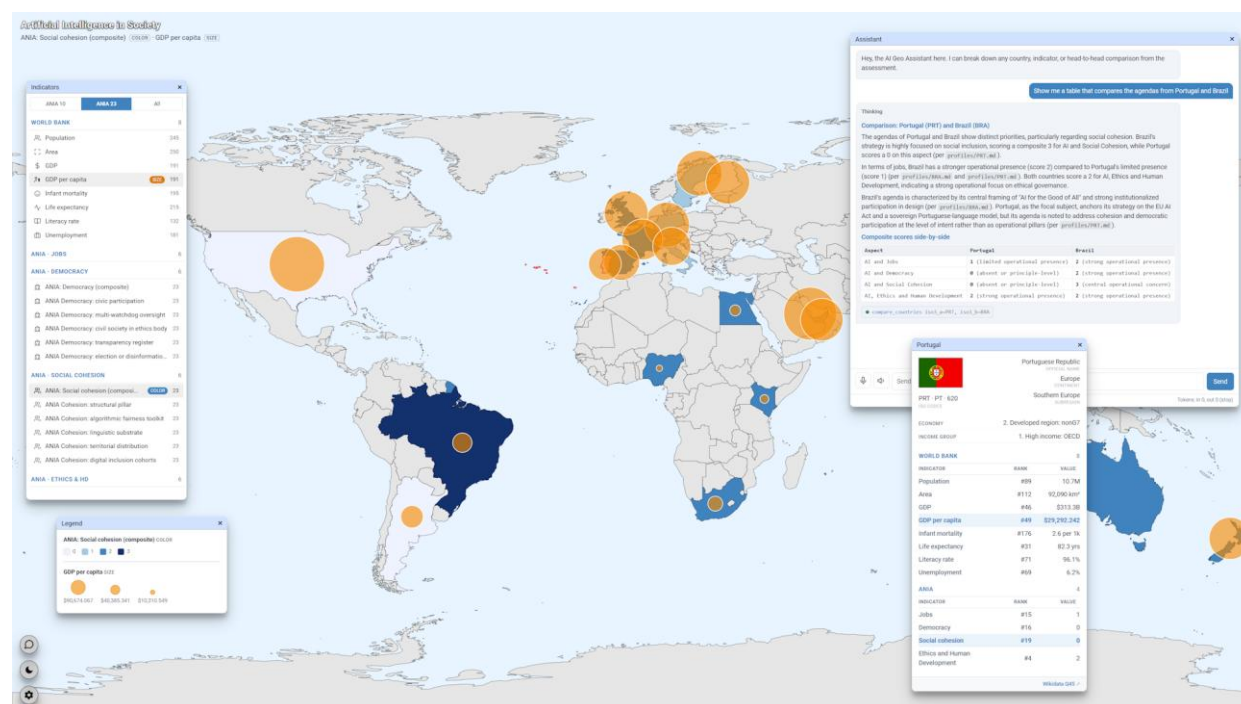


Figure C.7 - The ANIA Application interface.

C.6 Usage examples

The Application's chat surface was tested with twenty representative questions spread across four categories of interaction. Four examples below, one per category, illustrate the question types the Application is designed to answer and the corresponding chat-to-map binding the Assistant exercises in each case.

Per-country drill-down: “What does Portugal’s AI agenda say about social cohesion?”

The Assistant retrieves Portugal’s profile across the four ANIA categories, renders the per-pillar breakdown for Social Cohesion (showing the 0 composite with the single 1 from linguistic substrate), and refocuses the map on Portugal with the social-cohesion-composite layer active.

The interaction demonstrates the country-profile plus indicator-explainer tool pair and the focus-plus-layer-switch map binding that together let a reader audit any of the twenty-three countries in the corpus.

Indicator-specific: “How is ‘algorithmic fairness toolkit’ defined and which countries have one operational?” The Assistant returns the three-level rubric from §A.5, queries the dataset for countries scoring 2 on the corresponding pillar (Singapore is the only such country in the corpus), and switches the map layer to fairness-toolkit so the reader sees the global distribution immediately.

The interaction demonstrates the rubric-section retrieval and the query-and-highlight binding and surfaces the scope-exception that §A.6 documents (the Singaporean *AI Verify* reference).

Two-country comparison: “How do Brazil and France compare on social cohesion?”

The Assistant computes a structured comparison across the five social-cohesion sub-pillars, identifies the strengths each country brings (Brazil’s structural centrality and inclusion cohorts; France’s structural centrality and stronger overall composite), and highlights both countries on the map with the social-cohesion layer active.

The interaction demonstrates the compare-countries tool and the multi-country highlight binding and corresponds directly to the kind of paired-contrast reasoning the chosen-eight comparators in §4 of the main body are designed to make tractable.

Cross-cutting and pattern analysis: “What would Portugal need to do to lift its cohesion composite from 0 to 2?” The Assistant reads Portugal’s current row in the social-cohesion table, identifies the four pillars currently at 0 and the one at 1, draws a counterfactual scoring against the chosen-eight cohesion exemplars (Brazil, India, France, the Australia and New Zealand pair), and offers the reader the operational moves that would close the gap, anchored to the recommendation set in §5.2 of the main body.

The interaction demonstrates the highest-value composition the Application supports: the profile lookup, the comparison readout, and the recommendation-grounded synthesis, all chained from a single conversational question.